Ian Odundo

AI & Full-Stack Engineer

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AI and full-stack engineer with 6+ years building production web, mobile, USSD and payment systems across health, fintech and e-commerce. At AIDS Healthcare Foundation I build and ship global health information systems and AI services (fine-tuned multilingual NLP, LLM integration, RAG) used by about 2,000 people across 49 countries. Comfortable across the stack, from SQL Server data warehousing and payment APIs to polished, localized interfaces, with a strong user-centred design foundation.

WORK EXPERIENCE

Global Applications Developer · UI/UX

06/2024 – Present

AIDS Healthcare Foundation · Global HMIS, Nairobi, Kenya

- **AI / ML:** fine-tuned and self-host an XLM-RoBERTa multilingual sentiment API (Python, Hugging Face, FastAPI); integrated GPT-4o with prompt engineering and structured outputs; built RAG with embeddings and vector search; developed agent and tool-use workflows to automate data triage and reporting.
- **Data platform:** built the SQL Server data warehouse and led the SQL Server migration (T-SQL stored procedures, views, schema changes); built survey dashboards and data-visualization tools for online and offline use.
- **Platform engineering:** implemented OAuth2 single sign-on and a localization system serving 27 languages across 49 countries and about 1,300 facilities; built ASP.NET / C# REST APIs and Vue, React and Flutter interfaces.
- **Design and delivery:** gathered requirements with country teams and HMIS managers; produced wireframes, prototypes and specs in Figma; maintained a shared design system; shipped through Git and CI/CD. Use Claude Code to accelerate development.

Tech: Python, FastAPI, Hugging Face, GPT-4o, SQL Server, ASP.NET/C#, React, Vue.js, Flutter, OAuth2, Figma, Git

Software Engineer

11/2022 – 11/2023

Prevoir Solutions Informatiques (4C Group)

- Designed, built, tested and deployed Java services for **4C iNSight**, the group's telecoms and fintech platform (BI, fraud and revenue assurance, billing and a mobile-money payment gateway) deployed across 12 countries.
- Integrated card, mobile-money and bank payment APIs, including ISO 8583 messaging; built event-driven microservices on Kafka and RabbitMQ.
- Applied PCI-DSS controls: encryption, Spring Security, audit logging. Wrote JUnit unit and integration tests, GitLab CI/CD pipelines, Docker deployments, and technical documentation; built a React SPA; worked in an agile team with Jira and code review.

Tech: Java, Spring Boot, Spring Security, PostgreSQL, Kafka, RabbitMQ, ISO 8583, React, Docker, GitLab CI/CD, JUnit

Android Developer (Project Owner)

02/2022 – 08/2022

Doctors Live Telehealth · Volunteer, distributed team

- Owned the design and build of a telehealth Android app with WebRTC video consultations, real-time chat, Firebase (auth, Firestore, push) and M-Pesa payments.
- Handled patient data with privacy in mind; designed the UI in Figma; ran testing and Play Store release with a remote team using Git and agile practices.

Tech: Kotlin, Firebase, WebRTC, Retrofit, M-Pesa Daraja, Figma

Software Engineer

02/2021 – 06/2024

Lakeatts Systems Co. · Remote

- Designed, tested and deployed cross-platform products across Android, Flutter and USSD for clients including SDA Church, with push notifications via Firebase.
- Built Spring Boot / PostgreSQL backends and REST APIs; integrated M-Pesa Daraja, Equity Bank and I&M Bank payments.
- Administered Linux servers (Apache, Docker), hardened authentication and encryption, ran unit and UI tests with Git CI/CD, and supported clients.

Tech: Android (Java), Flutter, USSD, Spring Boot, PostgreSQL, Apache, Docker, Figma

CORE SKILLS

AI & Machine Learning

LLM APIs (GPT-4o) · Prompt engineering · RAG & vector search · AI agents · Model fine-tuning · XLM-RoBERTa · Hugging Face · Python · FastAPI · Claude Code

Web & Mobile

React · Vue.js · ASP.NET / C# · JavaScript · HTML5 / CSS3 · PHP / Laravel · Android (Kotlin, Java) · Flutter · i18n

Backend & Payments

Spring Boot · REST APIs · OAuth2 / SSO · M-Pesa Daraja · Card & bank APIs · ISO 8583 · Kafka · RabbitMQ · USSD

Data

SQL Server / T-SQL · PostgreSQL · MySQL · Data warehousing · Dashboards

DevOps & Security

Docker · GitLab CI/CD · Git · Linux · Apache · PCI-DSS · Secure coding · JUnit

Design & Leadership

Figma · Adobe XD · Illustrator · Design systems · Prototyping · Team lead · Requirements · Technical writing

EDUCATION

BSc Informatics & Computer Science

Strathmore University
07/2017 – 09/2021
Second Class Honours (Upper Division)

WORK EXPERIENCE (CONTINUED)

Software Developer · UI/UX Designer
01/2020 – 01/2021

Techxers Kenya Ltd · Nairobi, Kenya

- Joined through a work-based learning placement and stayed on as a freelance developer with the team, delivering Android, Flutter and web projects for clients including Neutron IT, CDL and Sahani.
- Built native Android apps in Java and Kotlin and cross-platform apps in Flutter, using Firebase for push notifications and backend services and consuming REST APIs.
- Developed Laravel / Lumen REST APIs on MySQL, including M-Pesa Daraja payment flows, and used Git for version control across client projects.
- Owned the UI/UX workflow: ran requirements sessions with clients, then produced wireframes and prototypes in Figma and Adobe XD, with assets in Illustrator and Photoshop.

Tech: Android (Java, Kotlin), Flutter, Firebase, Laravel, Lumen, PHP, MySQL, Figma, Adobe XD

Lead Software Developer
04/2019 – 09/2019

Thrive Solutions · Nairobi, Kenya

- Led a team of four developers building a mobile e-commerce platform end to end: allocated tasks, planned sprints, reviewed code and kept stakeholders informed.
- Built the Android app (Java) with Firebase push notifications, M-Pesa Daraja payments, and a product catalogue with image handling and search.
- Developed the Laravel / Lumen REST API on MySQL and a Vue.js admin web dashboard.
- Designed the app's UI/UX, managed source control in Git, and set up the servers and deployment.

Tech: Android (Java), Laravel, Lumen, PHP, MySQL, Vue.js, Firebase, M-Pesa Daraja

Freelance Developer & Designer
2018 – 2020

Self-employed · Nairobi, Kenya & Remote

- Delivered web, Android and branding projects for Helima Health, Bespoke Branding, Valentine's Cake House, Giftbox, Tatu City, Jambopay and SDA ECD, managing quoting, delivery and client communication end to end.
- Built responsive websites with Vue.js, PHP / Laravel, JavaScript, HTML5 / CSS3 and WordPress on MySQL, with SEO and performance optimization.
- Integrated M-Pesa, card and bank payments, and developed Android apps for client projects.
- Set up hosting, domains, cPanel and Docker environments for clients.
- Designed interfaces in Figma and Adobe XD and produced logos and visual identities in Illustrator and Photoshop.

Tech: Vue.js, PHP, Laravel, WordPress, MySQL, JavaScript, Docker, Android, Adobe XD, Illustrator, Photoshop, Figma

CERTIFICATES & ACHIEVEMENTS

IBM Cybersecurity (2020)
IBM Acclaim

MIT Covid-19 Africa Hackathon (2020)
Presented a tele-health solution

Red Bull Basement (2020)
Presented a ride-share system for Kenya

ICT Conference (2019)
Presented an online tendering system for academic institutions

Cyber Speak / The Hacker Village (2018)
Runner-up: ride-sharing system, 3-day hackathon

Certificate in Computer Applications & ICDL (2017)
Strathmore University

IT Essentials (2017)
Cisco Networking Academy

LANGUAGES

English
Native or bilingual

Kiswahili
Native or bilingual

French
Professional working

INTERESTS

- Artificial intelligence and machine learning
- UI/UX design
- Robotics

VOLUNTEERING

- Texas Cancer Centre